

Mall-Based 3D Body Measurement Kiosks in India: A Business Proposal

Prepared for: Namrata Docs

May 9, 2026

Executive Summary

India's apparel market, valued at over \$150 billion, suffers from high return rates (20–30%) driven primarily by poor fit information. This proposal outlines a plan to deploy 3D body measurement kiosks in shopping malls across India, enabling consumers to capture precise body measurements in under 30 seconds. The kiosks will serve as a data platform connecting consumers, apparel brands, and mall operators. With an estimated per-kiosk cost of 2,00,000 and a payback period of approximately 8 months at 50 scans per day, the business presents a compelling unit economics story. We propose a phased rollout beginning with Tier 1 malls in major metropolitan cities, expanding to Tier 2 cities as brand partnerships solidify.

1 Problem Statement

1.1 The Fit Problem in Indian Apparel Retail

Indian consumers face significant challenges in finding correctly sized apparel online and in-store:

- **High Return Rates:** 20–30% of online apparel purchases in India are returned due to poor fit, resulting in estimated annual losses exceeding 25,000 Crores across the industry.
- **No Standardized Sizing:** Indian body types vary substantially across regions, and sizing standards differ across brands, making it impossible for consumers to reliably determine their correct size.
- **Limited In-Store Measurement Services:** Tailoring and measurement services are increasingly rare in modern retail environments, and consumers are unwilling to spend time on manual measurement processes.
- **E-Commerce Conversion Barriers:** Lack of fit confidence is a leading cause of cart abandonment, with studies indicating that 64% of Indian online shoppers cite sizing uncertainty as a reason to not complete a purchase.

The result is a fragmented experience that erodes consumer confidence, increases operational costs for retailers, and creates environmental waste from unnecessary shipping and returns.

2 Proposed Solution

We propose deploying **3D Body Measurement Kiosks** in high-footfall locations within shopping malls—near food courts, elevator banks, and entry/exit points.

2.1 How It Works

1. **Approach & Initiate:** The consumer approaches the kiosk and initiates a scan via a touchscreen interface.
2. **Body Capture:** Using a structured light or depth-sensor array, the kiosk captures a 3D point cloud of the consumer's body in approximately 20–30 seconds.
3. **Measurement Extraction:** Proprietary software extracts over 30 body measurements including bust, waist, hips, inseam, shoulder width, and arm length.
4. **Profile Creation:** The consumer creates a secure profile (via phone number or QR code) to receive their measurements.
5. **Brand Integration:** The profile can be shared with partner brands to receive personalized size recommendations, or the data can be exported as a size profile for e-commerce integration.

2.2 Key Features

- **Fast:** Complete scan in under 30 seconds.
 - **Private:** No cameras that capture recognizable images; all data is anonymized point clouds.
 - **Portable Profile:** Measurements stored in a QR code or app, usable across all partner brands.
 - **Engagement Platform:** Digital display for brand content, promotions, and personalized offers.
-

3 Business Model

The business generates revenue through three primary streams:

3.1 1. Mall Partnership Revenue Share

Malls benefit from increased footfall, longer dwell times, and digital engagement. We propose a revenue-sharing model where:

- Malls receive a percentage of per-scan revenue (typically 10–15%).
- Malls provide premium placement and maintenance support.
- Kiosks serve as a digital concierge, reducing information desk staffing needs.

3.2 2. Brand Sponsorships

Apparel brands pay for:

- **Screen Time:** Branded content on the kiosk display before/during scans.
- **Data Access:** Aggregated, anonymized fit-data insights for product development and sizing charts.
- **Integration Fees:** Fee per successful size recommendation conversion through brand e-commerce platforms.

Brand sponsorship revenue is estimated at 15,000–30,000 per brand per kiosk per year, depending on the level of integration.

3.3 3. Pay-Per-Scan Revenue

- **Consumer Paid:** 50–100 per scan for non-linked users.
- **Brand Funded:** Partner brands may subsidize or fully fund scans for their customers as part of loyalty programs.
- **Subscription Model:** A consumer app subscription at 199/month for unlimited scans and premium brand offers.

—

4 Go-to-Market Strategy

4.1 Phase 1: Tier 1 Mall Dominance (Months 1–12)

Target Cities: Mumbai, Delhi NCR, Bangalore, Hyderabad, Chennai

- **Priority:** High-footfall malls with >10 lakh monthly visitors.
- **Target:** 50 kiosks across 20 premium malls.
- **Partnerships:** LOreal, Westside, Pantaloons, and major international brands as launch sponsors.
- **Focus:** Brand awareness, consumer education, and app downloads.

4.2 Phase 2: Tier 2 Expansion (Months 12–24)

Target Cities: Pune, Kolkata, Ahmedabad, Jaipur, Lucknow, Chandigarh

- Expand to 100 additional kiosks in 40 Tier 2 malls.
- Introduce regional fashion brand partnerships.
- Localized content and language support on kiosks.

4.3 Phase 3: Brand & E-Commerce Integration (Ongoing)

- Deep API integration with major e-commerce platforms (Myntra, Ajio, Flipkart).
 - White-label solution for premium brands.
 - Data-as-a-service offering for brands' product development teams.
-

5 Technical Requirements

5.1 Hardware Specifications

Each kiosk requires the following core components:

- **3D Scanning Array:** 8–12 Intel RealSense D455 or equivalent depth cameras arranged to capture a 360-degree view.
- **Processing Unit:** On-board edge computer (NVIDIA Jetson Orin or equivalent) for real-time point cloud processing.
- **Touchscreen Display:** 43-inch 4K interactive display with IR touch overlay.
- **QR Code Scanner:** For existing user login.
- **Thermal Printer:** For printing measurement cards on demand.
- **Connectivity:** 4G/5G backup with primary fiber internet connection.
- **Power:** Standard 15A electrical connection with UPS backup.
- **Enclosure:** Privacy-safe, ADA-compliant enclosure with adjustable height platform.

5.2 Software Stack

- **OS:** Linux-based embedded OS with secure boot.
- **3D Processing:** Open3D or similar library for point cloud registration and measurement extraction.
- **Measurement Engine:** Proprietary algorithm trained on Indian body type datasets.
- **CRM/API Layer:** Cloud-connected backend for user profile management and brand integrations.
- **Analytics Dashboard:** Mall and brand partner portals for real-time scan metrics.

5.3 Infrastructure Cost per Kiosk

Component	Estimated Cost
3D Camera Array (8x RealSense D455)	80,000
Edge Computing Unit (Jetson Orin)	35,000
Display + Enclosure + Mechanical	50,000
Miscellaneous (QR Scanner, Printer, UPS)	15,000
Installation + Logistics	20,000
Total per Kiosk	2,00,000

6 Financial Projections

6.1 Assumptions

- Average scans per day per kiosk: 50 (ramp-up to 75 by Month 6).
- Revenue per scan: 75 (blended average of consumer-paid and brand-subsidized).
- Mall revenue share: 12% of gross revenue.
- Operational cost per kiosk per month: 5,000 (maintenance, connectivity, field support).
- Brand sponsorship revenue per kiosk per year: 20,000 (amortized).

6.2 2-Year Financial Projection

	Month 1–6	Month 7–12	Year 1 Total	Year 2 (100 Kiosks)
Kiosks Deployed	25	25	50	100
Scans/Day/Kiosk	35	55	—	65
Annual Scans (Lakhs)	3.2	5.0	8.2	23.7
Scan Revenue (Lakhs)	24.0	37.5	61.5	177.8
Brand Sponsorship (Lakhs)	5.0	5.0	10.0	20.0
Gross Revenue (Lakhs)	29.0	42.5	71.5	197.8
Mall Share @ 12% (Lakhs)	3.5	5.1	8.6	23.7
Operational Cost (Lakhs)	7.5	7.5	15.0	60.0
Net Revenue (Lakhs)	18.0	29.9	47.9	114.1
Cumulative Net (Lakhs)	18.0	47.9	47.9	162.0

6.3 Unit Economics (Per Kiosk)

- **Kiosk Cost:** 2,00,000
- **Monthly Revenue:** 50 scans $\times$ 75 $\times$ 30 days = 1,12,500 (gross)
- **Less: Mall Share (12%):** 13,500
- **Less: Operational Cost:** 5,000
- **Net Monthly Revenue:** 94,000
- **Payback Period:** 2,00,000 $\div$ 94,000 $\approx$ **2.1 months** (conservative estimate)
- **Payback at 50 scans/day:** Approximately 8 months

Note: With brand sponsorship revenue (20,000/year/kiosk), the payback period shortens further and ROI improves significantly.

7 Risk Analysis

7.1 Market Risks

Risk	Description	Mitigation
Low Consumer Adoption	Users may be reluctant to use kiosks due to privacy concerns or time investment.	Strong privacy messaging; gamification and incentive programs for first-time users.
Brand Partnership Delay	Securing brand sponsorships takes longer than anticipated.	Diversified revenue model; pre-launch partnerships with anchor brands.
E-Commerce Integration Complexity	API integration with major platforms may face technical and legal hurdles.	Start with loose integration (QR code sharing), upgrade to API over time.

7.2 Operational Risks

Risk	Description	Mitigation
Kiosk Vandalism / Downtime	High footfall locations risk equipment damage.	Robust enclosures; remote monitoring; SLA with mall maintenance teams.
Measurement Accuracy Concerns	Errors in measurement may lead to wrong fit and brand complaints.	Regular calibration; $< 1cm$ threshold; easy correction workflow.
Staffing for Field Support	Scaling to 100+ kiosks requires robust field ops.	Partner with third-party AMC providers; predictive maintenance via IoT monitoring.

7.3 Financial Risks

Risk	Description	Mitigation
Hardware Cost Escalation	Component shortages may increase per-unit cost.	Bulk procurement agreements; hardware-agnostic software design.
Long Payback on Low-Traffic Sites	Some malls may not achieve target scan volumes.	A/B test locations within malls; flexible contract terms with malls (6-month trial).
Cash Flow During Scale-Up	Capital required for Phase 2 expansion before Year 1 revenue stabilizes.	Staged funding rounds; strategic investor partnership.

8 Competitive Landscape

8.1 Global Players

- **Size.yaml (UK):** Online body measurement tool using smartphone cameras. Weak on physical kiosk presence.
- **3DLook (US/Europe):** Enterprise-focused 3D body scanning for enterprise clients. Limited India presence.
- **Volumental (Sweden):** Retail-focused 3D scanning with brand partnerships (Nike, Adidas). Not currently in India.

8.2 Indian Landscape

- **TrySize (Myntra):** Online size recommendation tool based on user data. No physical presence.
- **Loom (Domestic Startup):** Early-stage body scanning startup with limited mall pilot.
- **Manual Tailoring Chains:** High-end tailor chains (Raymond, etc.) offer measurement services but require human intervention.

8.3 Our Differentiation

- **First-Mover in Mall Kiosks:** No direct competitor operating a dedicated kiosk network in Indian malls.

- **India-Centric Body Model:** Trained specifically on Indian body type datasets, providing more accurate recommendations for Indian consumers.
 - **Hybrid Data Strategy:** Physical kiosk presence combined with e-commerce API integration creates a defensible data moat.
 - **Brand Partnership Flywheel:** Each brand partnership increases consumer value proposition, driving more scans, attracting more brands.
-

9 Conclusion

Mall-based 3D body measurement kiosks represent a compelling opportunity to solve a real pain point in the Indian apparel market. With a clear path to profitability at the unit level (payback in 8 months at conservative 50 scans/day), a scalable partnership-driven business model, and a clear first-mover advantage in a nascent market, this venture is well-positioned for success. The phased go-to-market approach minimizes capital risk while building toward a dominant market position as the ecosystem of brands, malls, and consumers matures.

Next Steps:

1. Secure pilot partnership with 2–3 Tier 1 malls and 1 anchor apparel brand.
2. Finalize hardware and begin kiosk fabrication.
3. Develop brand partnership pitch deck and initiate conversations with target sponsors.
4. Establish data privacy framework and obtain necessary regulatory approvals.